Uttam Raj

uttam282005@gmail.com | linkedin.com/in/uttam282005 | github.com/uttam282005 | +91-6204784796

Education

National Institute of Technology, Agartala — Agartala, Tripura
Bachelor of Technology in Computational Mathematics, CGPA: 9.20/10 — *Aug. 2023 – Aug. 2027*

Technical Skills

Languages: Go, C++, C, TypeScript, Python, SQL
Backend & Systems: gRPC, PostgreSQL, Docker, POSIX Threads, Linux System Programming
Tools & Platforms: Git, GitHub Actions, Jest, Django, Linux, Vim, Opencode, Copilot
Concepts: Distributed Systems, Concurrency, Networking, REST APIs, CI/CD

Technical Projects

Limit Order Book Matching Engine | *C++20, Multithreading, Memory Pools* — Jun. 2026 | GitHub
- Architected a C++20 matching engine with custom memory pools and POD structs to eliminate heap allocations
- Enforced thread safety across concurrent insertions and cancellations via `std::shared_mutex` read-write locking
- Achieved 241K ops/sec throughput processing 400K market events with 5.79µs p50 and 142µs p99.9 tail latency

Distrack: Distributed Task Scheduler | *Go, gRPC, PostgreSQL, Docker* — Feb. 2026 | GitHub
- Engineered distributed task scheduler with Scheduler, Coordinator, and Worker services for delayed execution
- Designed gRPC heartbeats, round-robin dispatch, and transactional state updates in PostgreSQL
- Demonstrated 6.34 tasks/sec throughput and 5.9s avg E2E latency (p95 $\approx$ 11s) across 100 tasks and 6 workers

MapReduce Framework | *Go, RPC, Distributed Systems, Concurrency* — Oct. 2025 | GitHub
- Designed MapReduce framework in Go with Coordinator-Worker architecture communicating over RPC
- Orchestrated concurrent task scheduling with fault tolerance, auto-reassigning stalled tasks on 10s timeout
- Guaranteed crash-consistent outputs via atomic file renaming and FNV-1a hashing for intermediate partitioning

Custom Unix Shell | *C, Linux System Programming, POSIX* — Aug. 2024 | GitHub
- Developed Unix shell in C with built-in commands (echo, exit, type) and external command execution via PATH
- Enabled I/O redirection and multi-stage piping using `dup2` and process forking

Open Source Contributions

AboutCode (ScanCode.io & ScanCode-Toolkit) | *Python, Django, SPDX* — Jan. 2026 | GitHub
- Fixed URL-encoding of language filter values in resource list view, resolving broken filtering. (#2079)
- Added PlantUML license detection rule to scancode-toolkit, expanding OSS license coverage (#4666)

AWS Lambda Powertools (TypeScript) | *TypeScript, AWS Lambda, Jest* — Jan. 2025 | GitHub
- Refactored Metrics module to replace `EnvironmentVariablesService` with a cached `#envConfig`, cutting redundant env lookups (#4188)
- Fixed silent overwrite bug for default dimensions by emitting structured warnings on shared context misconfiguration (#4222, #4287)

Achievements

Top 1500 in **Google BigCode 2026**
National Semi-Finalist in **Flipkart GRiD 7.0** (2025)
Grand Finalist (Top 5) at national-level **Arjuna Hackathon**, organized by CITC, **NIT Agartala** (2024)
CodeChef: Global Rank **225** out of 25,000+ participants in **Starters 183** (2025) | Profile
Codeforces: Pupil (Peak Rating: **1237**) | **500+** problems solved | Profile

Leadership & Extracurricular

Technical Member – Developer & Coder's Club (DCC) — Aug. 2024 – Feb. 2025
NIT Agartala — *Agartala, Tripura*
- Built and deployed 3 event websites in React and Next.js for college technical events, handling live traffic
- Mentored 20+ juniors on web development fundamentals, Git workflows, and competitive programming onboarding